Where the non-dipolar phase lives in the Dipole-let domain, and what a cleaner can do about it

Results of the MCNO-QSM study

Technical report — October 2026

Abstract

We study the removal of the non-dipolar part ψ_N and the noise ε from a single-orientation local-field map $\psi = \psi_D + \psi_N + \varepsilon$ *before* susceptibility inversion, so that the streaking they cause in the eNDI reconstruction is reduced. Everything is expressed in the Dipole-let domain: $S = JM + 1$ coefficients $\mathfrak{D}_{j,m}\psi$ built from radial scales R_j and angular windows A_m around the magic cone Σ. We report (i) the geometry of ψ_N in that domain: noise is separable by scale (radial), the measured and modelled in-vivo residuals by angle ($|D|$), so one set of filter gains cannot serve both; (ii) that a field-domain Fourier network trained on noise sits at a measured information floor, and that none of six loss, architecture or data variants moves it; (iii) a zero-training *regime-gated passive Dipole-let filter* placed in front of that network, which lowers the eNDI error on the measured tissue residual from 71.7% to 58.6% (dirty 93.6%) without changing the noise-like cases; (iv) four negative results on the remaining failure, strong compact sources such as a calcification (amplitude gating, a passivity constraint, a source suppressor, plug-and-play refinement inside eNDI); and (v) a data-driven redesign of the decomposition, which improves the phase error but not the eNDI error. All numbers come from one seed, one training head (RC1) and one simulated test volume (Sim2); no in-vivo data were tested. Limits are stated next to each result and in section 8.

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1 Model, notation and aim

Model. Following [1], the measured local field (in ppm) is

$$\psi = \psi_D + \psi_N + \varepsilon, \qquad \psi_D = \mathrm{Re}\ \mathcal{F}^{-1}\big(D \cdot \mathcal{F}\chi\big), \qquad D(\boldsymbol{k}) = \tfrac{1}{3} - \frac{k_z^2}{\|\boldsymbol{k}\|_2^2}, \tag{1}$$

with the magic cone $\Sigma = \{D = 0\}$. ψ_N is the non-dipolar phase (microstructure, anisotropy, remnants) and ε the noise; m is the brain mask and w the eNDI weight (magnitude). The inversion is the authors' eNDI (`ndiCG`); the cleaner only acts on ψ. *Aim:* find an operator f with $f(\psi, m, w) \approx m \odot \psi_D$ such that the eNDI map of $f(\psi)$ is closer to χ than the eNDI map of $m \odot \psi$.

Dipole-lets. Radial filters R_j ($j = 0, \ldots, J-1$) and a low-pass residual R_{res}, with $\sum_j R_j + R_{\mathrm{res}} = 1$, and angular windows A_m ($m = 0, \ldots, M-1$), $\sum_m A_m = 1$, defined by thresholds $\delta_0 < \cdots < \delta_{M-2}$ on $|D|$:

$$\mathfrak{D}_{j,m}\psi = \mathcal{F}^{-1} R_j A_m \mathcal{F}\psi, \qquad \mathfrak{D}_{\mathrm{res}}\psi = \mathcal{F}^{-1} R_{\mathrm{res}} \mathcal{F}\psi, \qquad \psi = \sum_{j,m} \mathfrak{D}_{j,m}\psi + \mathfrak{D}_{\mathrm{res}}\psi, \tag{2}$$

$S = JM + 1$ segments, indexed by s. All results use $J = 3$, $M = 3$, $\delta = 0.05$, $\delta_1 = 1/3$, error-function windows, starlet scale $\sigma = 0.5$ voxel (so A_0 is the near-cone band $|D| < 0.05$, A_1 the band $0.05 < |D| < 1/3$, A_2 the rest). We also use the *split residual* $\mathfrak{D}_{\mathrm{res},m}\psi = \mathcal{F}^{-1} R_{\mathrm{res}} A_m \mathcal{F}\psi$, which replaces $\mathfrak{D}_{\mathrm{res}}$ by M pieces and gives $S' = JM + M = 12$ segments. A *passive Dipole-let filter* with gains $g \in [0,1]^S$ is

$$\mathcal{P}_g\psi = m \odot \sum_s g_s\, \mathfrak{D}_s\psi, \tag{3}$$

which can only attenuate segments (no gain above one, none negative).

Two regimes. We distinguish *noise-like* $\psi_N + \varepsilon$ (white, rough, as in the simulated Sim2) and *real-like* ψ_N (tissue residual, STI and off-diagonal terms measured or modelled on RC1).

2 Data and evaluation protocol

Data. RC1 is the FANSI challenge-1 head (160^3, 1.06 mm, TE 25 ms, B_0 2.89 T) with the COSMOS susceptibility, the measured tissue phase `phs_tissue` and the STI tensor. Sim2 is a $164 \times 205 \times 205$ simulation at 7 T (TE 8 ms) with a calcification (37 voxels below -0.5 ppm, χ down to -2.03 ppm) and a measured-like $\psi_N + \varepsilon$ (rms 0.0037 ppm, 84.7% of its energy above 0.2 cycles/mm). Sim2 is *only a test volume*: it is never used for training, selection or design choices; thresholds and gains are read from RC1. The clean target is always $y = m \odot \psi_D(\chi)$ with the correctly aligned kernel (the FANSI kernel is misaligned for odd matrix sizes).

Experiments (RC1, from the Dipole-lets notebook). Table 1 lists the fields on which methods are compared. exp7 is the only *measured* non-dipolar phase; exp8 and exp9 are modelled, and the filters of section 5 are fitted on noisy versions of exactly those two kinds of residual, so exp8 and exp9 are in-sample for the fit while exp7 is held out (same head, different residual).

Score. eNDI (`ndiCG`, $\alpha = 10^{-5}$, zero-padded to 192^3 for RC1 and 256^3 for Sim2), run for a *fixed* number of iterations, **25** (history index 25 of the NRMSE curve, i.e. 26 gradient steps), *not* with the automatic stop. The score is NRMSE $= 100\,\|m \odot (\hat{\chi} - \chi)\| / \|m \odot \chi\|$. Fixed iterations are used because the dirty fields diverge after 4–17 iterations and the stop fires at very different iterations for different fields, which confounds comparisons. Phase NRMSE ($\|m \odot (\hat{\psi} - \psi_D)\| / \|m \odot \psi_D\|$) is reported where useful but is not the criterion (section 3.2). Oracle (eNDI of y): 35.8% (RC1), 43.1% (Sim2).

exp1	$\psi = \psi_D(\chi_{\mathrm{COSMOS}})$ masked (noiseless: any change is damage)
exp3	ψ_D with complex noise at SNR 40
exp5, exp6	ψ_D plus cylinders ($r = 8$, ± 0.5 ppm) / spheres ($r = 4$, ± 0.33 ppm) added to the phase: *strong localized sources*
exp7	measured `phs_tissue` (the real residual $\psi = \psi_D + \psi_N$)
exp8, exp9	the STI field $p_{33} - (p_{13} + p_{23})$ / the clean field minus the off-diagonal term, $\psi_D - p_Q$
Sim2 real	`Frequency.nii`: the measured-like $\psi_N + \varepsilon$, calcification included
Sim2 forward	$\psi_D(\chi_{\mathrm{Sim2}})$ plus noise (SNR 100, calibrated, 40), plus phase jumps, plus χ spheres

Table 1: Experiments.

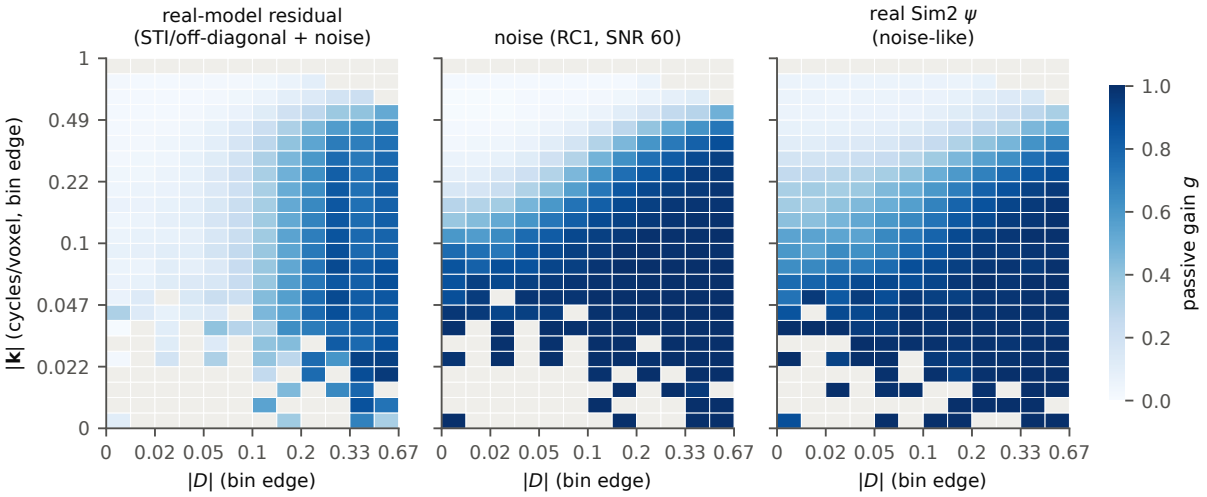


Figure 1: Best passive gain per cell of the $(\|\boldsymbol{k}\|, |D|)$ plane (light grey: cell without energy). Left: modelled in-vivo residuals (STI and off-diagonal terms plus noise, RC1): the gain depends on $|D|$ and hardly on $\|\boldsymbol{k}\|$. Middle: white noise on RC1: the gain depends on $\|\boldsymbol{k}\|$ (about 1 below 0.1 cycles/voxel, falling to $\lesssim 0.1$ above 0.4 for $|D| < 0.1$) and weakly on $|D|$. Right: the noise-like ψ of the real Sim2.

Rules followed. Success criteria were written down before each run (they are quoted below); a negative result is recorded as negative; every number is a single seed and a single volume unless a spread is given.

3 Geometry of the non-dipolar phase in the Dipole-let domain

3.1 Angular versus radial separability

For each regime we accumulate, on a grid of 24 radial bins ($\|\boldsymbol{k}\|$ in cycles/voxel) $\times$ 12 bins of $|D|$, the cell energies $E_b = \sum|\hat{y}|^2$, $E_{bd} = \sum \mathrm{Re}(\hat{y}\,\overline{\hat{\psi}})$, $E_d = \sum|\hat{\psi}|^2$ ($\hat{y}, \hat{\psi}$ the transforms of the clean and the dirty masked field). The best passive scalar gain of a cell is $g = \mathrm{clip}(E_{bd}/E_d, 0, 1)$. Figure 1 shows g for the three regimes. The two regimes are separated by different variables:

- **Real-like ψ_N is angular.** The share of the energy that lies close to Σ is far higher than for the clean field (table 2); the local signal-to-residual ratio $|\hat{y}|^2/|\hat{\psi}_N|^2$ is -13 to -17 dB in $|D| < 0.05$ at *every* $\|\boldsymbol{k}\|$ (even below 0.03 cycles/mm), about -9 dB in $0.05 < |D| < 0.1$, and $+4$ to $+11$ dB above 0.2. The optimal gain rises from ≈ 0.1 at $|D| < 0.05$ through 0.4 at 0.15 and 0.7 at 0.27 to 0.85 at 0.33, independent of scale. This is the premise of the Dipole-lets paper (non-dipolar content in the near-cone bands at every scale).

- **Noise is radial.** For Sim2's noise-like ψ the local signal-to-residual ratio changes with $\|\boldsymbol{k}\|$ (+25 dB below 0.03, 0 dB at 0.2–0.3, −12 dB above 0.4 cycles/mm) and only 10–15 dB along $|D|$; the three radial scales of the decomposition resolve only $\|\boldsymbol{k}\| \gtrsim 0.1$ cycles/voxel and the residual carries 78.6% of the true-field energy.

	$\lvert D\rvert < 0.05$	$\lvert D\rvert < 0.1$
clean field y (RC1)	0.21	0.88
noise ε	12	24
real tissue residual	13	26
STI term	20	37
off-diagonal term	11	21

Table 2: Share (%) of the energy that lies in the near-cone region, RC1.

Remark 1. *The cone share alone does not discriminate noise from real residuals (white noise puts* 12% *in* $|D| < 0.05$*, the real residual* 13%*); what does is the* scale dependence*: the local excess of cone energy over its white-noise expectation falls by a factor* ~ 10 *from the finest to the coarsest scale for noise and is flat across scales for the real residuals (section 5).*

3.2 An information floor in the field domain

Within each cell the best *scalar* filter fitted on the volume itself (an optimistic oracle) leaves 27.27% phase error on Sim2; the trained network leaves 27.16%, and a best-fit linear filter with one gain per segment leaves 26.44% against the network's 26.26%. The network is at, and in the near-cone cells below, that floor: for noise-like $\psi_N + \varepsilon$ nothing is left to extract with a linear-in-cell or per-segment filter, and a weighted-L2 loss of any kind has the same minimiser (the conditional mean). Two consequences matter below. (a) Re-weighting the loss cannot help (confirmed, section 4). (b) **Phase NRMSE does not predict eNDI error**: the tissue-only network reduces the phase error on Sim2 (42.2 → 37.4%) yet its eNDI error is 60.6% against 52.6% for the dirty field; four different decompositions in section 7 differ by 3 points of phase error and < 0.9 points of eNDI error. All claims are therefore made in eNDI.

4 The field-domain network (stage B) and its plateau

Stage B is the Dipole-let Fourier network of the companion report: a residual FNO whose spectral weights are tied to the $S = 10$ Dipole-let segments (complex channel mixing per segment, modulated by a small network of $D(\boldsymbol{k})$, $\|\boldsymbol{k}\|$, $\hat{\boldsymbol{k}}$), 67,305 parameters, $\hat{\psi}_D = m \odot (\psi - \widehat{\psi}_N)$. The recipe that works on Sim2 is *noise*: RC1 patches of 96^3 with complex noise at SNR 40–200 (plus 20% identity examples, *noise_clean*, so that clean fields are left alone), trained on one head. We call this network B. Table 3 summarizes a long series: neither a loss that re-weights bands or Dipole-let segments (with measured eNDI sensitivities g_s), nor a local convolution, a U-Net branch, 85 additional anatomies, or a training family modelled on the calcification changes the result beyond 0.5 points. Capacity is not the limit: the same network removes phase-additive localized artifacts of its own making to 97% on fresh draws, and the train and held-out noise ratios are equal (0.466 and 0.449, no overfitting). Mixing structured families into the recipe is *worse* on Sim2 (49.05), a negative task interference. The calcification family, the one designed for Sim2's failure, does not recover the field in the 3 mm shell around the source (relative error 1.02, gain 0.27), because the *input* there keeps only 32% of the true field.

	auto-stop eNDI	25 fixed it.
dirty (masked)	52.60	53.94
low-pass $\sigma = 0.5$ (tuned on the target)	49.03	–
all nine artifact families (mix, 3 seeds)	49.05 ± 0.19	50.0
noise (3 seeds)	$\mathbf{46.96 \pm 0.04}$	47.28
noise + band-normalized spectral loss	47.47	–
noise + local dilated conv in each block (3 seeds)	46.43 ± 0.45	–
noise + U-Net branch (3 seeds; branch inert)	46.67 ± 0.22	–
noise_clean (identity examples)	–	**47.31**
noise_clean + 85 AHEAD anatomies	–	47.71
noise_clean + Dipole-let sensitivity-weighted loss	–	47.27
noise_clean + calcification-shell family	–	47.36
oracle	44.22	43.10

Table 3: eNDI NRMSE (%) on the real Sim2 (single volume). The first column is the early series with the automatic stop (masked), the second the fixed-iteration protocol used from here on. Every variant of the noise recipe lies within ~ 0.5 points of it.

5 A regime-gated passive Dipole-let filter in front of the network

5.1 Linear Dipole-let filters on real-like residuals

With no training, one real gain per segment fitted by least squares on RC1's modelled residuals (STI and off-diagonal terms with noise at SNR 100, *never* the measured tissue) and applied to the measured tissue residual exp7 already removes most of the damage (table 4). Splitting the residual by angle, $\mathfrak{D}_{\text{res}} \to \mathfrak{D}_{\text{res},m}$, is worth 1.4–2.7 eNDI points on exp7–9 (the phase gain was 2.5–4 points); in the phase proxy more radial scales ($J = 4, 5$) give the same linear gain as the split and saturate at $J = 6$, and in eNDI the 15- and 16-segment filters are within 0.3 of the 12-segment one; none of these changes the result on noise-like fields. A 12-parameter linear filter

filter (gains from the other two real types)	exp7	exp8	exp9
dirty	93.6	67.7	83.3
network B (noise_clean)	71.4	60.9	75.9
cone notch $\lvert D\rvert < 0.15$ on the dirty field	64.8	52.8	60.7
10 segments ($J = 3, M = 3$)	62.2	55.6	59.5
12 segments (split residual)	**59.7**	**54.2**	**56.8**
15 segments ($J = 4$ + split)	59.6	54.2	56.7
16 segments ($J = 5, M = 3$)	59.8	54.4	57.1

Table 4: Zero-training linear Dipole-let filters, eNDI NRMSE (%) at 25 iterations. Fitted gains of one real type are applied to the others (exp7 is never fitted).

beats the trained network B on the real residuals by 12, 7 and 19 points, and the trained network behaves on them like a radial Wiener filter. The price: the gains of the *real* regime applied to Sim2's noise-like ψ give 57–58% (worse than dirty, 53.9), while the gains fitted on noise give 49.2 and 69–71 on exp7/exp9: choosing the wrong regime costs 8–12 points either way.

5.2 The regime map and the gated filter

The regime must be read from the input. Let G_ρ be a Gaussian of $\rho = 6$ voxels and, for the detail scale j,

$$\phi_j(\boldsymbol{x}) = \log \frac{G_\rho * (\mathfrak{D}_{j,0}\psi)^2}{G_\rho * \sum_{m\geq 1} (\mathfrak{D}_{j,m}\psi)^2} - \log \mu_j, \qquad \mu_j = \frac{\|R_j A_0\|^2}{\sum_{m\geq 1} \|R_j A_m\|^2}, \tag{4}$$

the local log excess of near-cone energy over its white-noise expectation ($\phi_j = 0$ for white noise). Medians of $\exp\phi_j$ over 96^3 patches: noise 0.15–0.78 at the finest scale falling to 0.06–0.09 at the coarsest; the real Sim2 ψ 0.47/0.26/0.13; tissue, STI and off-diagonal residuals flat across scales (0.50/0.43/0.42, 0.34/0.36/0.38, 0.24/0.26/0.28). The coarsest detail scale separates the regimes with a margin of ~ 1.4. We use it only:

$$r(\boldsymbol{x}) = \mathrm{sigm}\big(6\,\phi_2(\boldsymbol{x}) + c\big), \qquad c = -6\ln 0.22 = 9.0848, \tag{5}$$

($r = 0.5$ at an excess of 0.22; $r \approx 0.00$ on noise, 0.05 on the mean of the real Sim2, 0.93/0.89/0.78 on exp7/exp8/exp9). **Stage A** removes the real-regime correction where the gate is on:

$$\mathcal{A}\psi = m \odot \Big(\psi - r \odot \sum_s (1 - g_s)\,\mathfrak{D}_s\psi\Big),\ g = \arg\min_{g\in[0,1]^{S'}} \sum_f \Big\| m \odot \Big(\sum_s g_s \mathfrak{D}_s \psi^{(f)} - \psi_D\Big)\Big\|^2, \tag{6}$$

with the 12 segments of the split residual, f running over the two modelled residual types (STI and off-diagonal), $\psi^{(f)}$ their noisy versions on RC1 at SNR 100 and the bound $g \in [0, 1]$ (passive filter, eq. (3)). The fitted gains are

$$g = (0,\ 0,\ 0.40,\ 0,\ 0.74,\ 1.0,\ 0,\ 0.77,\ 0.83,\ 0,\ 0.67,\ 0.92) \tag{7}$$

for $(\mathfrak{D}_{0,0}, \mathfrak{D}_{0,1}, \mathfrak{D}_{0,2}, \mathfrak{D}_{1,0}, \ldots, \mathfrak{D}_{\mathrm{res},2})$: the near-cone band $|D| < 0.05$ is removed at every scale and in the residual, the band $0.05 < |D| < 1/3$ is attenuated (removed entirely at the finest scale), the far band is kept. A first fit without the bound has gains in $[-1.4, 1.3]$ and costs ≈ 1 point less on exp7–9 but amplifies strong sources (section 6); the bound costs about one point and halves that damage. **The cascade** is $\widehat{\psi}_D = B(\mathcal{A}\psi)$: stage A (no training; gains and gate read from RC1 only) followed by the unchanged network B. In the noise regime ($r \to 0$) it reduces to B.

5.3 Results

Table 5 and fig. 2 compare the dirty field, B alone, the joint network that learns the gains and the gate jointly with the data (12 segments, trained on the mixture of noise and modelled residuals) and the cascade. Criteria fixed beforehand: exp7 ≤ 63; no harm on noise-like data (real Sim2 ≤ 47.8, forward Sim2 fields within 0.5 of B, exp1 ≤ 36.8, exp3 within 1.0); exp8 and exp9 at least 5 points better than B. **Verdict against the criteria.** Met: exp7 ($58.6 \leq 63$); exp8/exp9 (9 and 16 points better than B); forward Sim2 fields (largest +0.35), exp1 (36.11), exp3 (+0.03). *Not* met: real Sim2 ($48.72 > 47.8$, +1.24 over B). The soft gate and a hard gate ($r > 0.5$) agree within 0.9 everywhere.

6 The remaining failure: strong compact sources

Localization. The real-Sim2 loss (+1.24 over B for the cascade) comes from the neighbourhood of the calcification: for the unconstrained stage A alone, which was the version localized (the passive one was not re-localized), it sits entirely within 15 mm of it. In the field domain that stage A *lowers* the error everywhere (42.2 → 39.6% phase NRMSE; 0.65× inside the gated voxels far from the calcification) and raises it only in the 3–8 mm ring around it (1.21×). In eNDI (laptop run, history index 24, dirty 53.47): applying the unconstrained stage A everywhere 54.91; only farther than 15 mm from the calcification 53.41 (no harm); only within 15 mm 54.95; only where $r > 0.5$ 55.04. The same happens on the RC1 strong sources exp5/exp6 (stage A alone 198/76 with the unconstrained gains against 94/36 dirty). The filter zeros near-cone content of true fields of compact sources, and the cone-excess gate does not exclude them.

	dirty	B	joint net	stage A alone	**cascade** $B \circ \mathcal{A}$
exp7 (measured tissue)	93.60	71.69	**56.58**	61.37	58.63
exp8 (STI)	67.69	61.15	**50.40**	52.79	52.35
exp9 (off-diagonal)	83.33	76.33	**53.60**	62.09	60.46
Sim2 real	53.94	**47.48**	48.62	54.62	48.72
Sim2 fwd SNR 100	61.48	**46.69**	46.97	61.11	46.86
Sim2 fwd SNR cal.	67.52	**47.44**	47.90	68.35	47.71
Sim2 fwd SNR 40	121.45	**52.13**	55.37	113.68	52.48
Sim2 fwd jumps	66.81	**48.43**	50.58	67.00	48.57
Sim2 fwd χ spheres	70.57	**51.68**	51.87	69.63	51.76
exp1 (noiseless)	**35.76**	36.11	37.54	35.77	36.11
exp3 (SNR 40)	61.15	**43.02**	43.98	60.69	43.05
exp5 (cylinders)	**94.14**	116.13	169.87	105.67	113.56
exp6 (phase spheres)	**35.83**	46.60	68.59	65.15	62.95

Table 5: eNDI NRMSE (%) at 25 fixed iterations. The cascade equals B within 0.35 on all forward-simulated fields, exp1 and exp3, lowers the measured residual (exp7) by 13 points relative to B and exp8/exp9 by 9/16; the joint network is better on exp7–9 but costs 0.2–3.2 points on noise-like data. Real Sim2: B is 47.48 here and 47.31 in the trainer's own output (a different outside-the-mask convention).

stage A alone, eNDI@25 (laptop, index 24)	exp5	exp6	exp7	exp8	exp9	Sim2 real
dirty	93.6	36.0	92.8	67.7	83.3	53.47
unconstrained gains	198.1	75.6	60.2	51.9	61.1	54.91
+ amplitude suppressor, $\sigma = 6$ voxels	157.4	51.0	60.2	–	–	54.91
passive gains $g \in [0, 1]$	104.1	63.2	61.2	52.7	61.9	54.40
passive + suppressor, $\sigma = 2.5$, 0.06–0.10 ppm	124.8	45.2	61.2	52.7	61.9	54.37

Table 6: Attempts to protect strong sources. Suppressor: the gate is multiplied by $1 - \mathrm{smoothstep}\big((\ell - \ell_0)/(\ell_1 - \ell_0)\big)$, ℓ the local rms of the field; thresholds read from RC1 only (real residuals and noise stay below 0.06 ppm at $\sigma = 2.5$, exp5/exp6 reach 0.41/0.16 at p99.9). (The exp8/exp9 entries of the first two rows are from the server table, history index 25.) None meets the exp5, exp6 or Sim2 criteria; the suppressor does nothing on Sim2 because the compact calcification does not reach the threshold after smoothing (thresholds were *not* retuned on Sim2).

Four remedies, all negative. Criteria for stage A alone were fixed before each run (exp5 ≤ 99, exp6 ≤ 40, exp7 ≤ 63, exp8 ≤ 58, Sim2 real $\leq$ *dirty* $+ 0.3$).

1. **Amplitude gating** (two scales of smoothing): fails; suppressing stage A at strong sources also removes its benefit on the cylinders of exp5 (they are real non-dipolar phase).
2. **Passivity** ($g \in [0, 1]$): keeps the real-residual gain (exp7 61.2 against 60.2) and halves the strong-source damage, but not enough (exp5 104, exp6 63, Sim2 +0.9).
3. **Plug-and-play refinement inside eNDI** (table 7): the cascade operator $C = B \circ \mathcal{A}$ is applied to the *residual* of the current estimate, $r_k = \psi - m \odot K x_k$, and the result is added back, $b_k = m \odot K x_k + C(r_k)$, then eNDI continues from x_k; round 0 with 6 steps and four rounds of 5 steps (26 in all). It is worse than the one-pass cascade on *every* row, and fails real Sim2 (51.97 > 47.8), exp7 (70.7 > 63) and all forward fields (+4.4 to +10.2 over B).
4. **Data-driven decomposition** (section 7): improves the phase proxy, not eNDI, and is worse on strong sources.

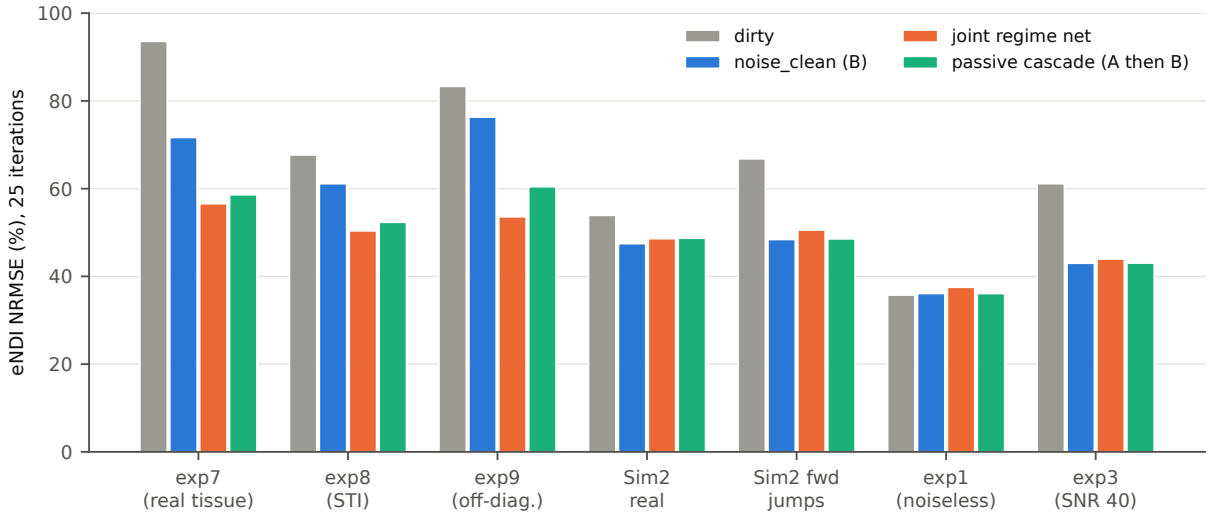


Figure 2: Selected rows of table 5 (eNDI NRMSE, lower is better): dirty field, network B, joint regime network, passive cascade. The cascade follows B on the noise-like cases and follows the joint network on the real residuals.

	exp7	exp8	exp9	Sim2 real	snr100	snr40	exp1	exp3
one-pass cascade	58.63	52.35	60.46	48.72	46.86	52.48	36.11	43.05
plug-and-play, $C = B \circ \mathcal{A}$	70.73	55.72	70.39	51.97	51.12	62.33	38.13	47.46
plug-and-play, $C = B$	77.17	62.00	82.50	50.01	49.41	61.52	36.59	48.02
plug-and-play, $C = \mathcal{A}$	72.00	56.02	70.87	53.19	54.79	91.60	37.56	52.83

Table 7: Plug-and-play residual refinement (eNDI NRMSE, 26 steps). A plausible but untested reason: the cleaners were never trained on a low-amplitude, noise-dominated residual, and they are scale-sensitive.

7 A data-driven second-generation decomposition

The cell statistics of section 3 also allow *designing* the decomposition. A Dipole-let segment is a radial window times an angular band, so the natural design space is: contiguous radial groups, each with its own contiguous partition of $|D|$. For a candidate partition $\mathcal{Q}$ with passive gains $g_q = \mathrm{clip}(\sum_{c\in q} E_{bd} / \sum_{c\in q} E_d, 0, 1)$ the residual energy is $\sum_q \sum_{c\in q}(E_b - 2g_q E_{bd} + g_q^2 E_d)$, minimized exactly by dynamic programming for $S = 2, \ldots, 20$ segments.

- **Real regime.** Identity: 61.2%; per-cell floor 36.9%; best partitions 38.9 ($S = 4$), 38.2 (6), 38.0 (8), 37.7 (12), 37.35 (20): flat beyond ~ 8 segments. The best angular edges are $|D| \in \{0.10, 0.15, 0.20, 0.27\}$ instead of the hand-set $\{0.05, 1/3\}$; the $S = 6$ partition has $\|\boldsymbol{k}\| < 0.40$ cycles/voxel with edges $\{0.10, 0.20, 0.27\}$ and $\|\boldsymbol{k}\| \geq 0.40$ with edge $\{0.27\}$.
- **Noise regime.** Identity 43.4%, floor 24.4%, $S = 12$ gives 24.9% with radial edges at $0.15, 0.27, 0.40, 0.49$ cycles/voxel, $|D|$ edges 0.15–0.2 only at higher $\|\boldsymbol{k}\|$ and none for the coarse remainder. The (noise-like) Sim2 ψ agrees (edges $0.08, 0.22, 0.33, 0.49$).

So the two regimes want *different* decompositions (few angular bands and little radial structure for real-like ψ_N; about five radial groups and almost no angular bands at low $\|\boldsymbol{k}\|$ for noise); the hand-set 12 segments are a compromise between them. Table 8 evaluates the real-regime designs with smooth windows (gains fitted on the modelled residuals, tested on the held-out tissue), in phase and in eNDI. The derived layouts are 2.8–3.1 points better in phase NRMSE but all four decompositions lie within 0.9 points in eNDI on exp7: *the decomposition is not the bottleneck*. Six

	S	phase NRMSE		eNDI NRMSE (ungated, 25 it.)				
		tissue	STI+n	exp7	exp8	exp9	exp5	exp6
dirty	–	73.6	52.3	93.6	67.7	83.3	94.1	35.8
current Dipole-let (split residual)	12	44.17	35.85	59.09	53.34	56.11	110.8	70.9
derived, real regime	6	41.42	32.83	58.57	53.72	56.37	179.1	80.5
derived, real regime	8	41.24	32.69	58.48	53.65	56.35	181.6	80.6
derived, real regime	12	41.05	32.46	58.23	53.40	55.87	178.5	79.1
derived, noise regime	12	48.36	39.95	–	–	–	–	–

Table 8: Data-driven decompositions against the current one. Criteria set beforehand for $S = 6$ against the current 12: exp7 at least 1.5 points better (got 0.52), exp8/exp9 not worse by more than 0.5 (met), exp5/exp6 not worse by more than 2 (failed: +68, +9.6). Not adopted.

data-derived segments do as well as twelve hand-set ones on exp7–9, a simplification, but their wide near-cone bands zero the cone at every scale and damage strong sources more.

8 Limitations

- **One seed, one training head, one test volume.** Spreads are only known for the noise recipe (three seeds, ±0.04). The cascade and the filters have no seed (no training), but depend on RC1's single head for the gains and the gate.
- **exp8 and exp9 are in-sample for the fit** (the same modelled residuals, noisy versions, same head); exp7 is the only held-out real residual and shares its anatomy with the fit data.
- **No in-vivo data were tested.** The gate was calibrated on one head and one measured residual; its margin on the coarsest scale is only ~ 1.4 (12 patches per set).
- **Strong compact sources are unsolved** (the Sim2 calcification: +1.24 over B; exp6 63 against 46.6). The input of the 3 mm shell keeps only 32% of the true field, so part of this is information, not method.
- **Phase-based proxies are not eNDI.** Design choices made with the linear proxy (gains, decomposition) transferred only partly.
- The eNDI score depends on the iteration count; 25 fixed iterations is a choice justified by divergence of the dirty fields, not a tuned optimum. The oracle (35.8/43.1) is not a ceiling because the mask removes signal that any masked field loses.

9 Conclusions

1. In the Dipole-let domain the non-dipolar phase is **angular** (near-cone bands at every scale) for real-like residuals and **radial** (fine scales, any $|D|$) for noise. A single filter cannot serve both; the regime can be read from the input by the scale dependence of the cone excess of the finest-to-coarsest detail bands.
2. For noise-like $\psi_N + \varepsilon$ the field-domain network is at a measured floor and no loss, architecture, data or family variant tested changes it.
3. A **zero-training, regime-gated passive Dipole-let filter** followed by the unchanged noise network lowers the eNDI error of the measured tissue residual from 71.7% to 58.6%, and of the modelled residuals by 9–16 points, without changing noise-like cases (largest change +0.35); it costs +1.24 on the real Sim2 because of the calcification.

4. The split of the coarse residual by angle is the only change to the hand-set decomposition that matters for real residuals (1.4–2.7 eNDI points); more scales, other edges, or a data-driven redesign do not improve eNDI.
5. Strong compact sources are the open problem; amplitude gating, passivity, a source suppressor, plug-and-play refinement and a redesigned decomposition did not solve it.

Reproducibility

Results are in `CLAUDE.md` of the project (sections 9l–9ax, with the criteria written before each run) and come from `scripts/cascade_gate.py`, `scripts/pnp_endi.py`, `scripts/design_dipolelet2.py`, `scripts/check_design_endi.py`, `scripts/check_passive_filter.py`, `scripts/check_passive_strong.py`, `scripts/gate_suite2.py`, `scripts/analyze_streak_kspace.py` and the S4b training scripts; tables 5 and 7 are `runs/cascade_passive/table.txt` and `runs/pnp/table.txt`. Figures: `figures/make_results_figures.py`.

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